reverence and disrespect for received wisdom.

Physics is an exact science, and the existing body of knowledge, while incomplete, can't simply be shrugged aside. Feynman acquired a formidable grasp of the accepted principles of physics at a very young age, and he chose to work almost entirely on conventional problems. He was not the sort of genius to beaver away in isolation in a backwater of the discipline and to stumble across the profoundly new. His special talent was to approach essentially mainstream topics in an idiosyncratic way. This meant eschewing existing formalisms and developing his own highly intuitive approach. Whereas most theoretical physicists rely on careful mathematical calculation to provide a guide and a crutch to take them into unfamiliar territory, Feynman's attitude was almost cavalier. You get the impression that he could read nature like a book and simply report on what he found, without the tedium of complex analysis.

Indeed, in pursuing his interests in this manner Feynman displayed a healthy contempt for rigorous formalisms. It is hard to convey the depth of genius that is necessary to work like this. Theoretical physics is one of the toughest intellectual exercises, combining abstract concepts that defy visualization with extreme mathematical complexity. Only by adopting the highest standards of mental discipline can most physicists make progress. Yet Feynman appeared to ride roughshod over this strict code of practice and pluck new results like ready-made fruit from the Tree of Knowledge.

The Feynman style owed a great deal to the personality of the man. In his professional and private life he seemed to treat the world as a hugely entertaining game. The physical universe presented him with a fascinating series of puzzles and challenges, and so did his social environment. A lifelong prankster, he treated authority and the academic establishment with the same sort of disrespect he showed for stuffy mathematical formalism. Never one to suffer fools gladly, he broke the rules whenever he found them arbitrary or absurd. His autobiographical writings contain amusing stories of Feynman outwitting the atom-bomb security services during the war, Feynman cracking safes, Feynman disarming women with outrageously bold behavior. He treated his Nobel Prize, awarded for his work on QED, in a similar take-it-or-leave-it manner.

Alongside this distaste for formality, Feynman had a fascination with the quirky and obscure. Many will remember his obsession with the long-lost country of Tuva in Central Asia, captured so delightfully in a documentary film made near the time of his death. His other passions included playing the bongo drums, painting, frequenting strip clubs, and deciphering Mayan texts.

Feynman himself did much to cultivate his distinctive persona. Although